

Tracking Consciousness with Median Nerve Stimulation-based BCI during General Anesthesia

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Through EEG and BCI



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and Biology Society

Accidental Awareness during General Anesthesia (AAGA)



Yuko Shimizu

Monitors to avoid AAGA

Traditional clinical signs are unreliable indicators of anesthesia depth [1], which leads to the use of brain activity monitors.

Awareness still occurs with current monitors [1, 2, 3, 4, 5] : indices were developed from adult cohorts [5], might be falsely increased with EMG activity [3, 4] and are less accurate with age [5]



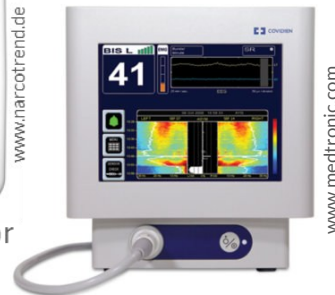
NeuroSENSE monitor



SedLine monitor



Narcotrend monitor



Bispectral Index monitor

[1] Bruhn et al. Anesth, 2000.

[2] Myles et al. Lancet, 2004.

[3] Sebel et al. Anesth. Analg. 2004.

[4] Aho et al. Acta Anaesthesiol Scan, 2012.

[5] Laferrière-Langlois et al. Anesth. Analg., 2024.

AAGA patients try to move

Realised she was awake when she should not be. She tried to get attention, tried to talk and move but was unable. Heard voices and noise from surgical instruments.

Suddenly noticed endotracheal intubation, but felt no discomfort. Was scared. Tried to move but found it impossible. Heard somebody say “she is not asleep”.

Suddenly saw a big, round lamp. Tried to move his arms and scream, but was unable. Had impression that surgeon was about to finish. Soon was unconscious again.

Paralysed before she became unconscious. Realised something was wrong, although did not understand what. Tried to move but was unable. Repeated explanations were required. Anxiety gradually disappeared over 2 weeks.

During AAGA, patients **try to move** to alert the anesthesiologist [6], yet it is not always possible due to neuromuscular blocking agents, which actually increase this risk [7].

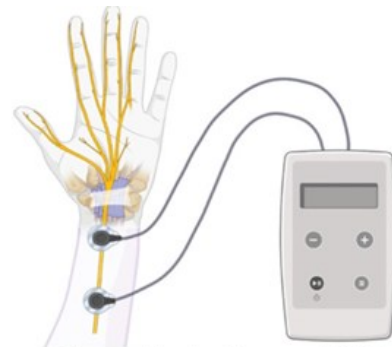
[6] Sandin et al, The Lancet, 2000.

[7] Pandit et al, Br J Anaesth, 2013.

Median Nerve Stimulation (MNS)

SOLUTION: Actively analyze the motor cortex with EEG responses to **Median Nerve Stimulation** (MNS) [8].

- Painless electrical stimulation (5 to 15 mA)
- Provides a trigger



Thomas Hendry, Neuromod, 2024

MNS-based Brain Computer Interface (BCI)

$$ERD/ERS\% = \frac{\overline{x^2} - \overline{BL^2}}{\overline{BL^2}} \times 100$$

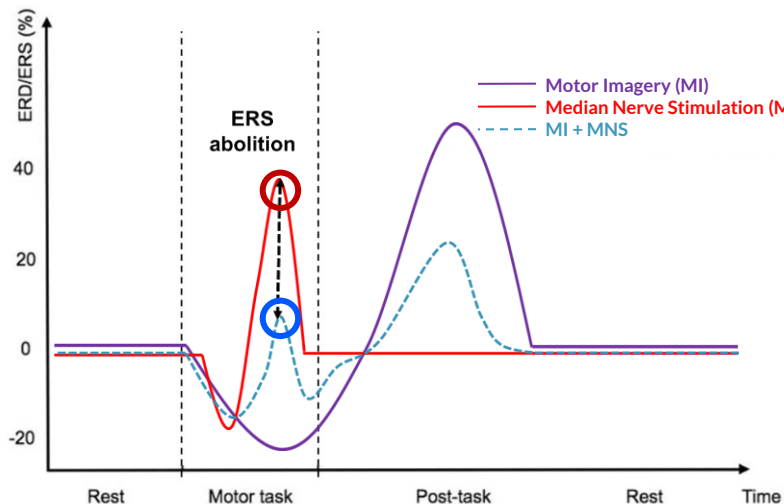
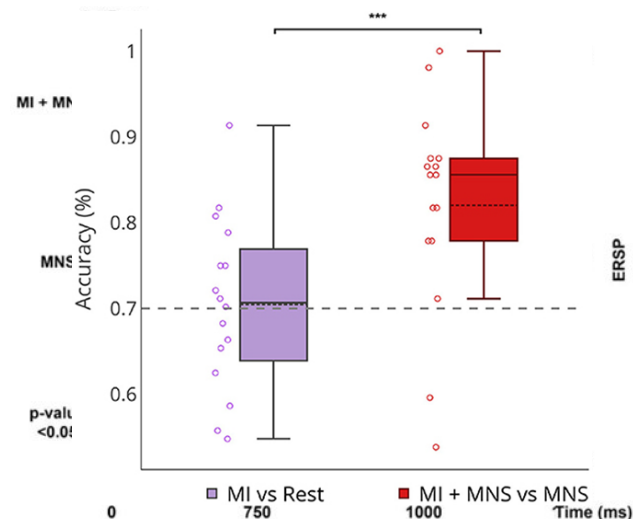


Illustration of ERD/ERS% [9] induced by MI, MNS and MNS+MI, in the alpha and beta frequency bands [8]



Distribution of average classification accuracies (n = 16), obtained by a linear classifier with p-value < 0.001 [8] (grand average, n = 16) in 8-30 Hz [8]

MNS sensorimotor modulations change with concomitant MI [10, 11], leading to **better MI detection accuracy (+12%)** [8].

[8] Rimbert et al, Front. Neurosci, 2019.

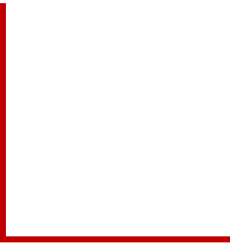
[9] Pfurtscheller and Lopes da Silva, Clin Neuroph, 1999.

[10] Cheron et al, Electroenceph. Clin. Neuroph., 1987.

[11] Salenius et al, NeuroImage, 1996.

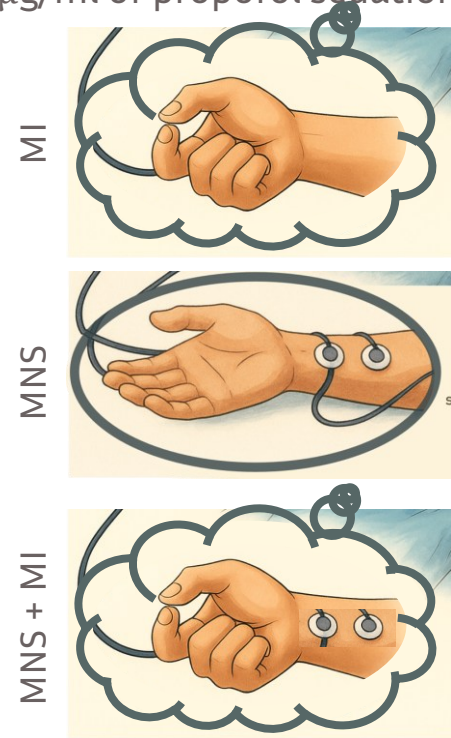
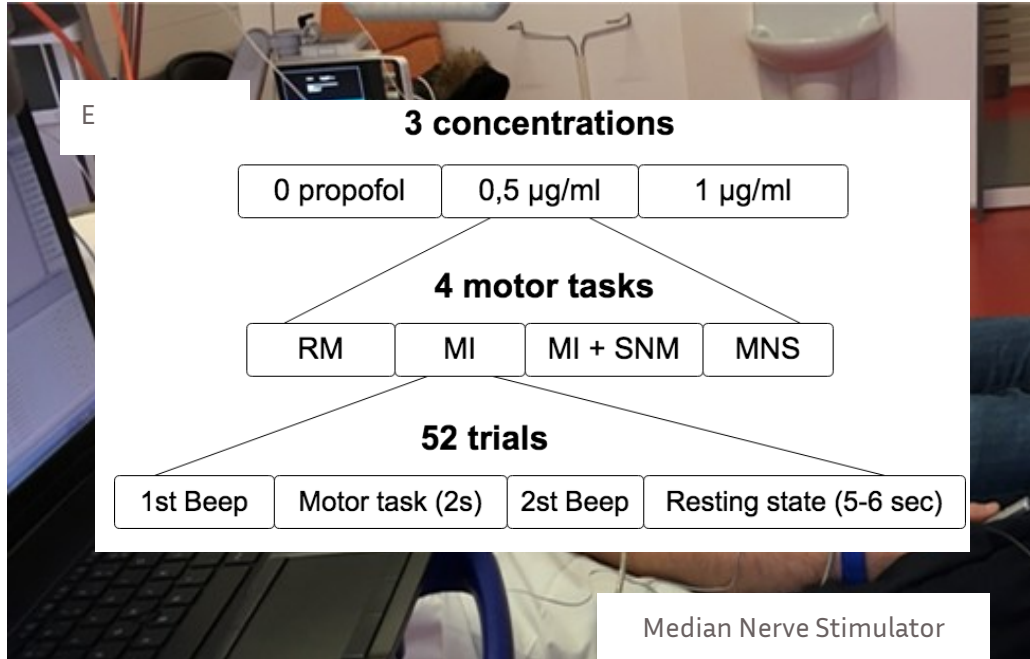


I. Median Nerve Stimulation-based BCI to detect motor intentions **under sedation**

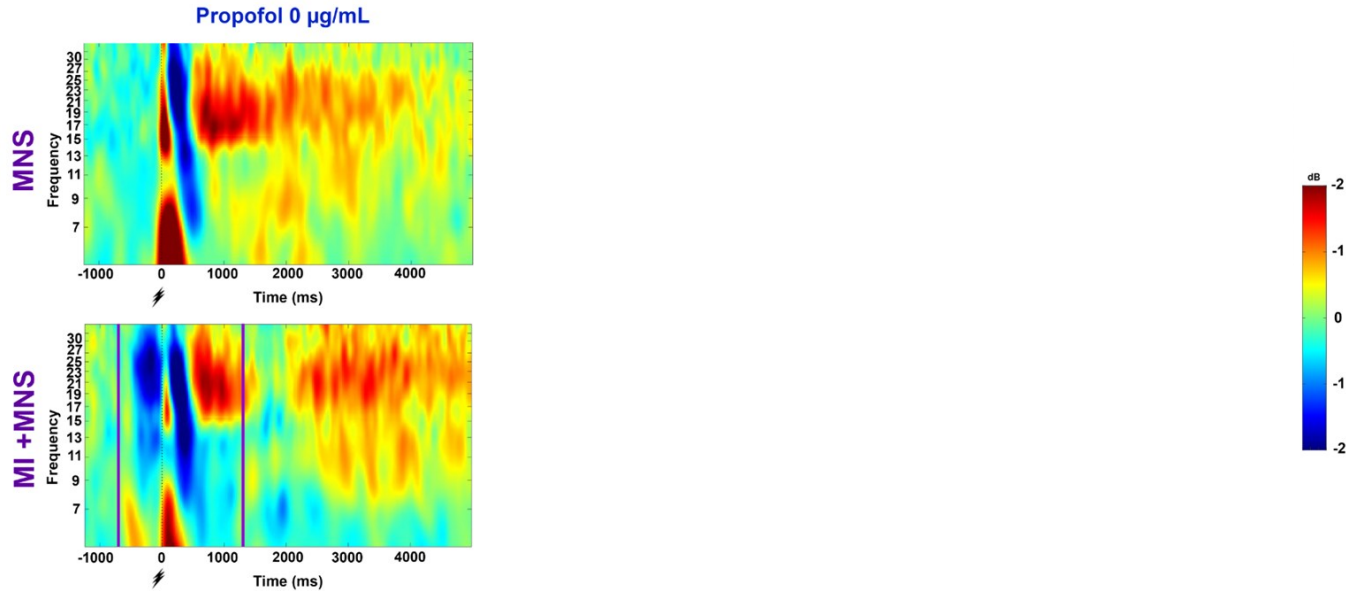


Motor Imagery (MI) detection at 3 sedation levels

Dataset: 31 healthy subjects, 51 trials of MI, MNS or MNS+MI at 0, 0.5, 1 $\mu\text{g/ml}$ of propofol sedation [12].



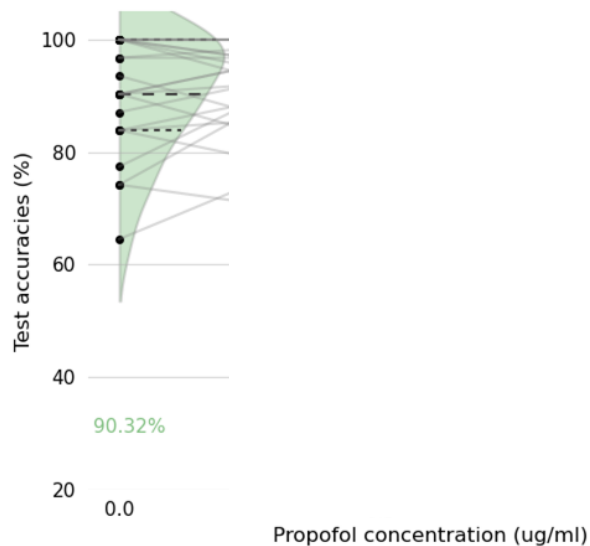
MI+MNS vs MNS under sedation



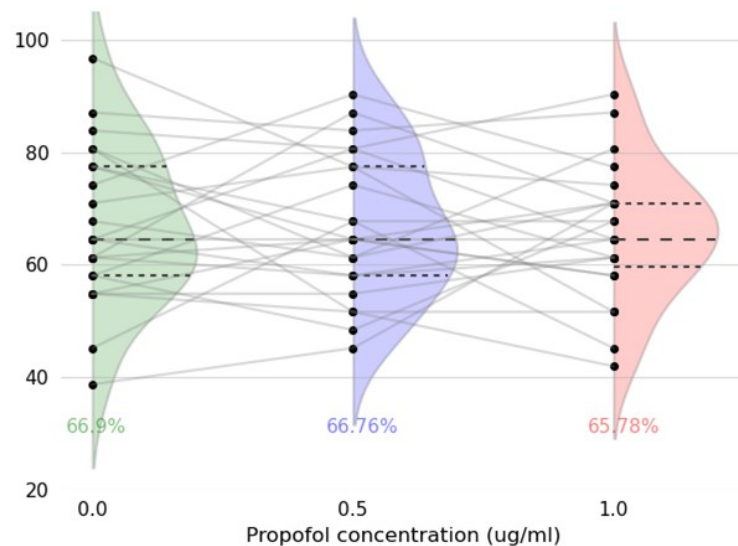
Time frequency maps (grand average, $n = 16$) in 8-30 Hz, for electrode C3 at different propofol concentrations [8]

Detection of motor imagery under sedation

MNS + MI vs MNS



MI vs REST



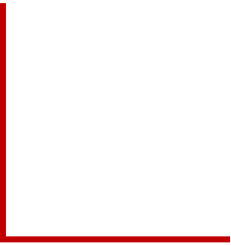
Summary: MI detection under sedation

- ERD/ERS% patterns are still visible under sedation.
- MI detection accuracies are still higher using MNS (+22% at 1 $\mu\text{g/ml}$ of propofol sedation).
- Loss of consciousness typically occurs at propofol concentrations around 1.5–2.0 $\mu\text{g/ml}$ [13].

How do ERD/ERS responses behave at **higher propofol concentrations**?



II. Median Nerve Stimulation-based BCI to monitor **anesthesia**



MNS during general anesthesia

Dataset: 14 patients, 8 females, 50 ± 14.3 years old [14].

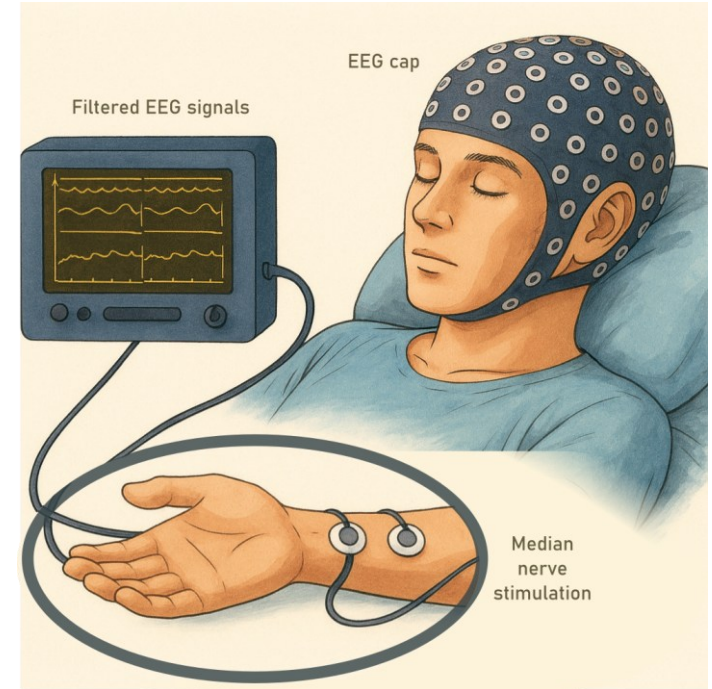
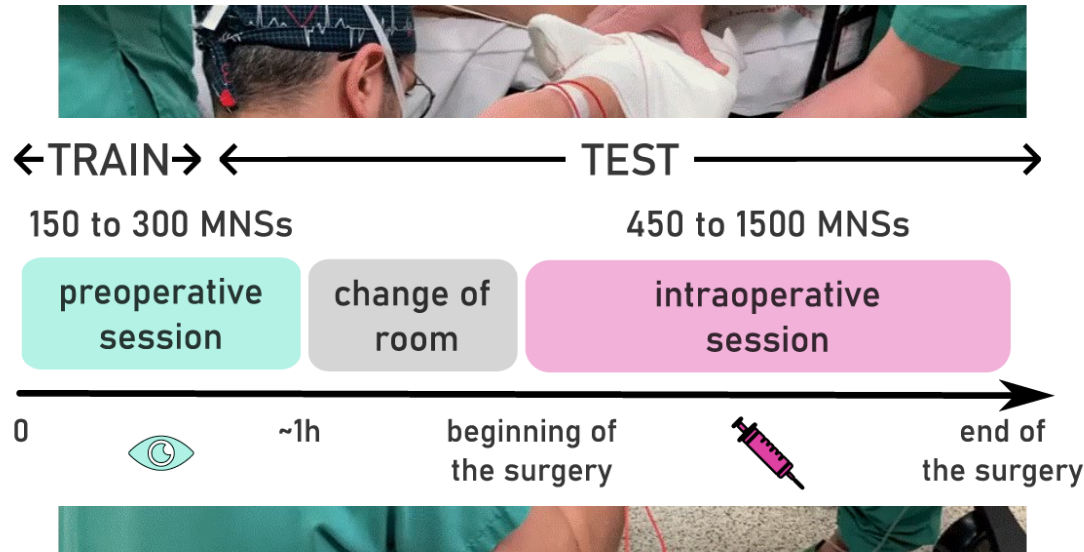


Figure inspired by Sigl and Chamoun, J Clin Monit, 1994

EEG recording difficulties in a clinical environment

Inclusion of subjects:

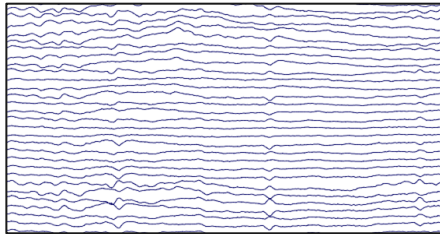


- consentment
- position during surgery
- unmaintained surgery
- dysfunctional equipment

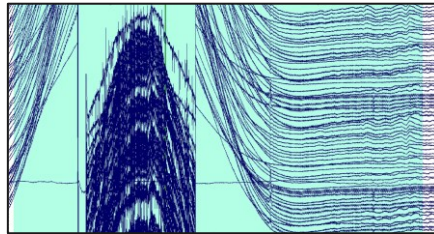


Interferences:

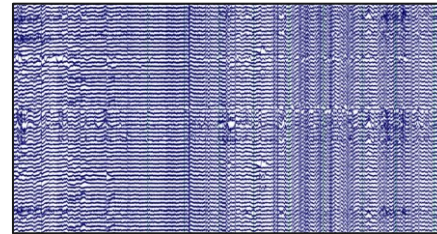
- hot air blankets [15]
- electrocautery [16]
- individual socket
- line noise (50 Hz or 60 Hz)
- movement



CLEAN



ELECTROCAUTERY



LINE

www.keewell.com.cn

www.meditec.ca

www.radiamed.ca

[15] Hemmerling et al. Anesth. Analg. 2002.

[16] Hight et al. Can. J. Anesth. 2021

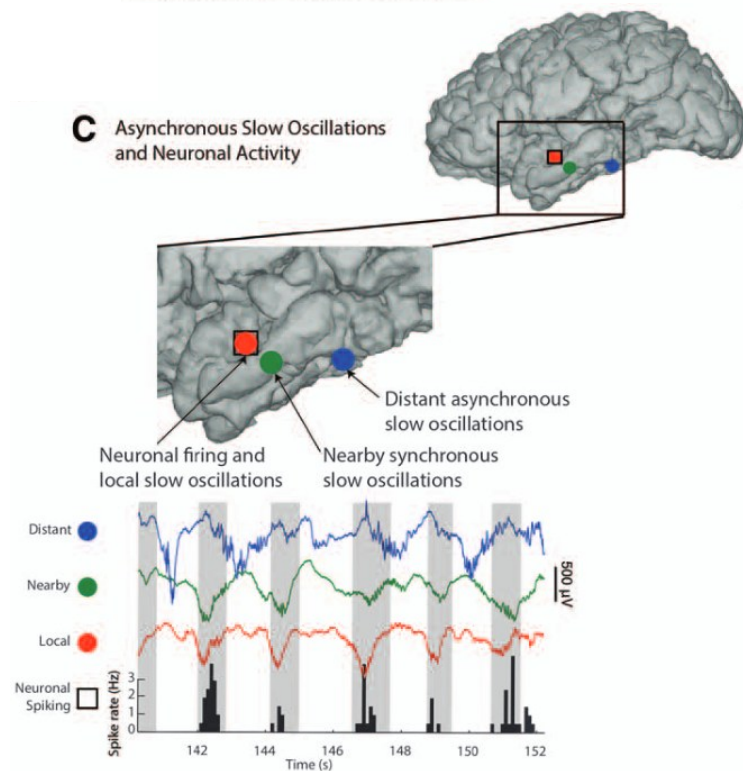
Propofol-induced oscillatory patterns

Alpha oscillations within circuits linking the thalamus and the frontal cortex.

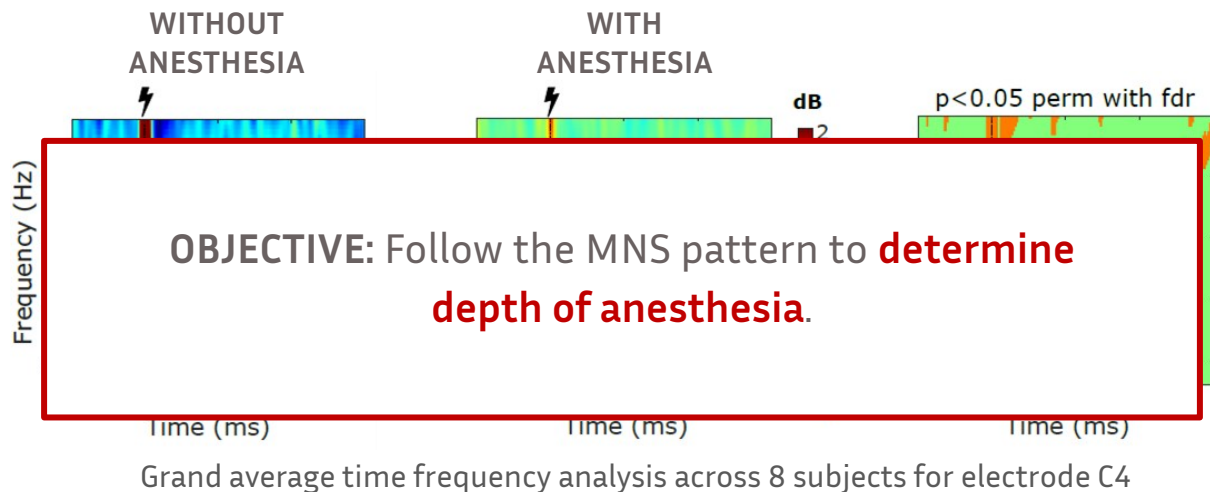
Intracortical fragmentation as communication through spiking activity is restricted to local areas.

Coherent alpha rhythms might **prevent thalamo-cortical communication**, while incoherent slow oscillations **impair intracortical communication** [17].

Thalamocortical Circuits Underlying Propofol-Induced Alpha Oscillations



MNS patterns during general anesthesia



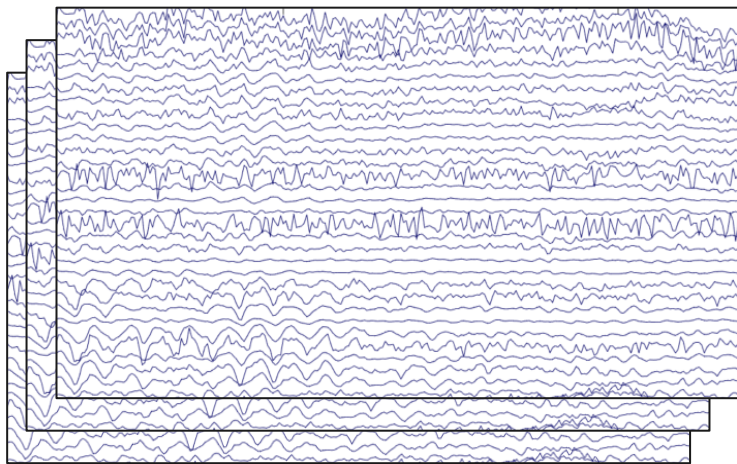
MNS sensorimotor modulations are **variable with the depth of anesthesia** [9].

Awake & anesthetized states

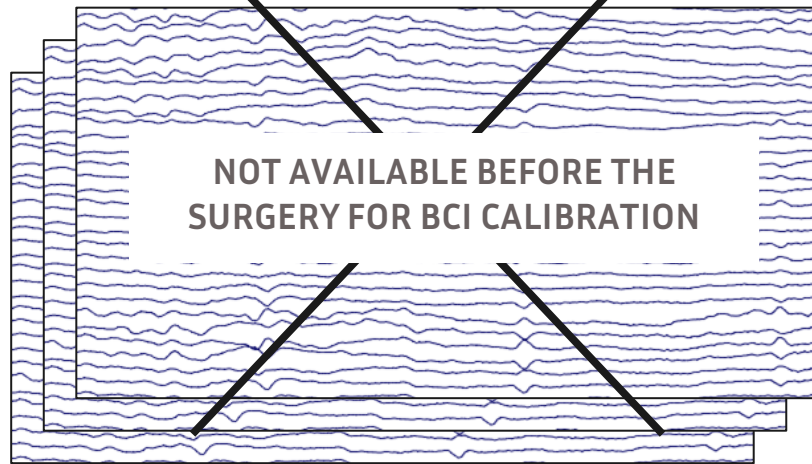
GOAL: Classify brain state as either **MNS-awake** or **MNS-anesthesia** using a BCI.



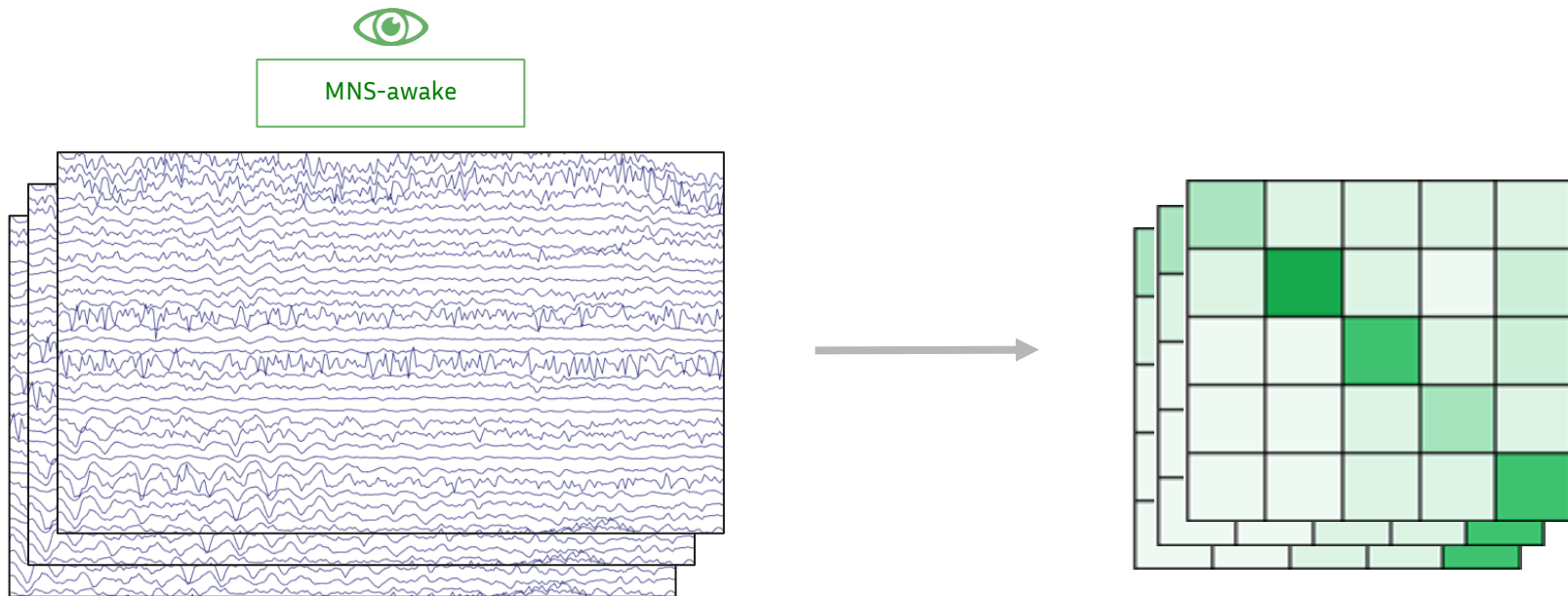
MNS-awake



MNS-anesthesia



One-Class Riemannian Minimum Distance to the Mean (OCR-MDM)

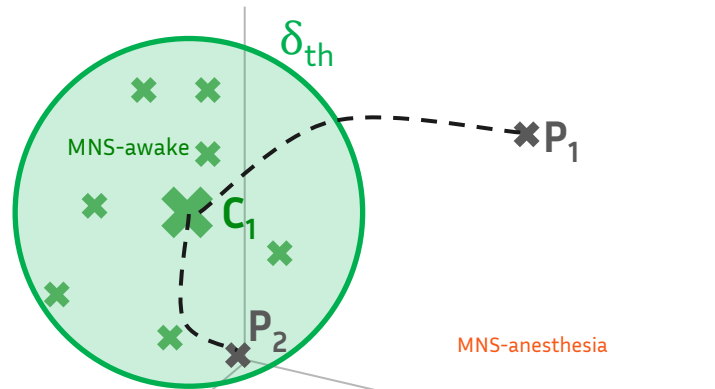
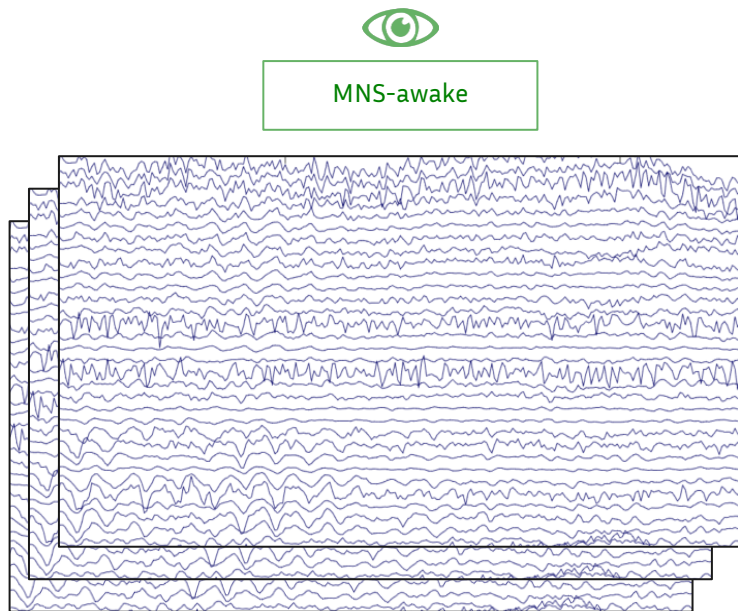


[17] Congedo et al. 8th Inter. BCI Conf. 2019.

[18] Barachant et al. IEEE Trans. Biomed. Eng. 2012.

[19] Marissens Cueva et al. 5th Int. Neuroergonomics Conf, 2024.

One-Class Riemannian Minimum Distance to the Mean (OCR-MDM)



New covariance matrix $\mathbf{P}$ of class k :

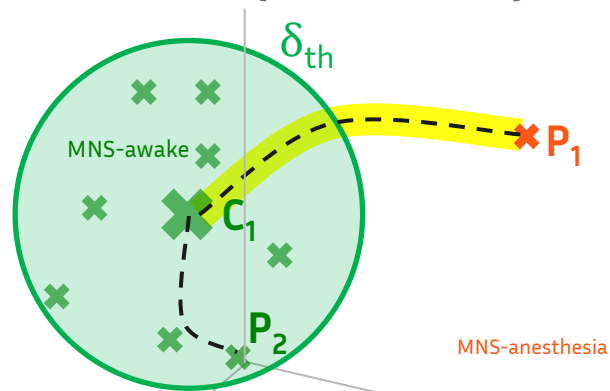
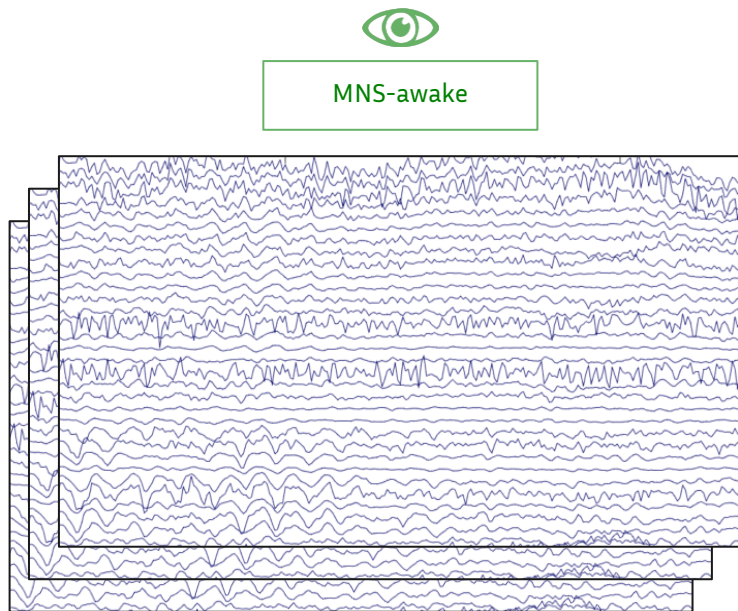
$k = C_1$ if $\delta_R(\mathbf{P}, \mathbf{P}_i) \leq \delta_{th}$ else C_2
 with δ_R the Riemannian distance:
 $\delta_R(\mathbf{P}, \mathbf{P}_i) = \|\log(\mathbf{P}^{-1/2} \mathbf{P}_i \mathbf{P}^{-1/2})\|_F$

[17] Congedo et al. 8th Inter. BCI Conf. 2019.

[18] Barachant et al. IEEE Trans. Biomed. Eng. 2012.

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One-Class Riemannian Minimum Distance to the Mean (OCR-MDM)



New covariance matrix $\mathbf{P}$ of class k :

$$k = C_1 \text{ if } \delta_R(\mathbf{P}, \mathbf{P}_i) \leq \delta_{th} \text{ else } C_2$$

with δ_R the Riemannian distance:

$$\delta_R(\mathbf{P}, \mathbf{P}_i) = \|\log(\mathbf{P}^{-1/2} \mathbf{P}_i \mathbf{P}^{-1/2})\|_F$$

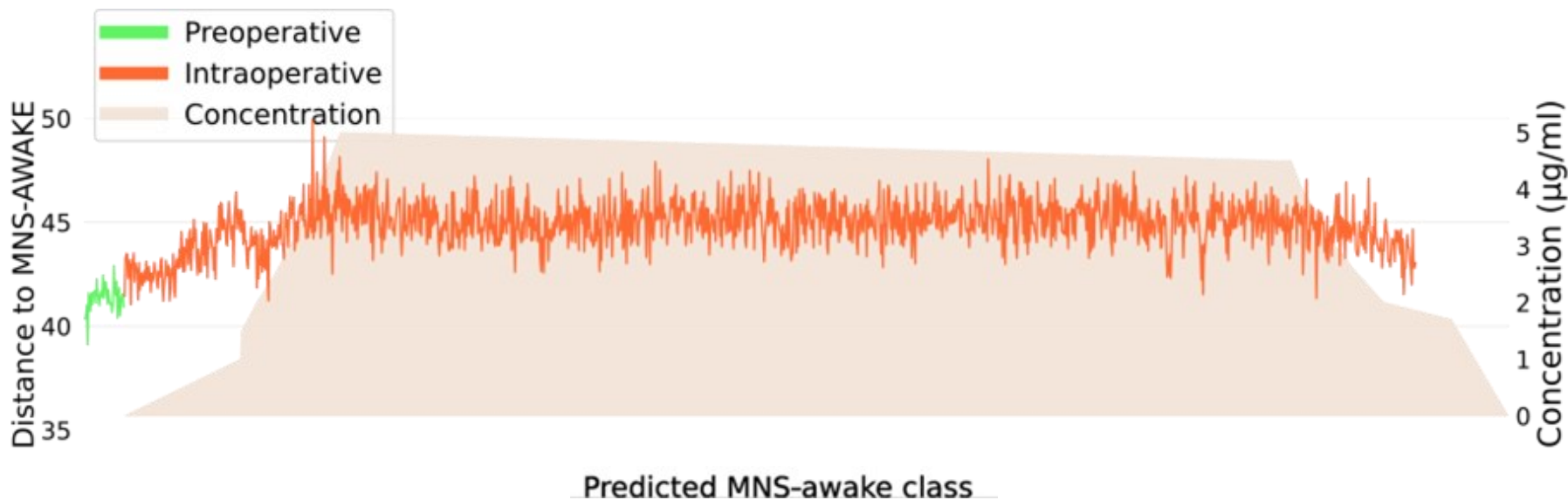
[17] Congedo et al. 8th Inter. BCI Conf. 2019.

[18] Barachant et al. IEEE Trans. Biomed. Eng. 2012.

[19] Marissens Cueva et al. 5th Int Neuroerg Conf, 2024.

Towards a quantitative anesthesia index

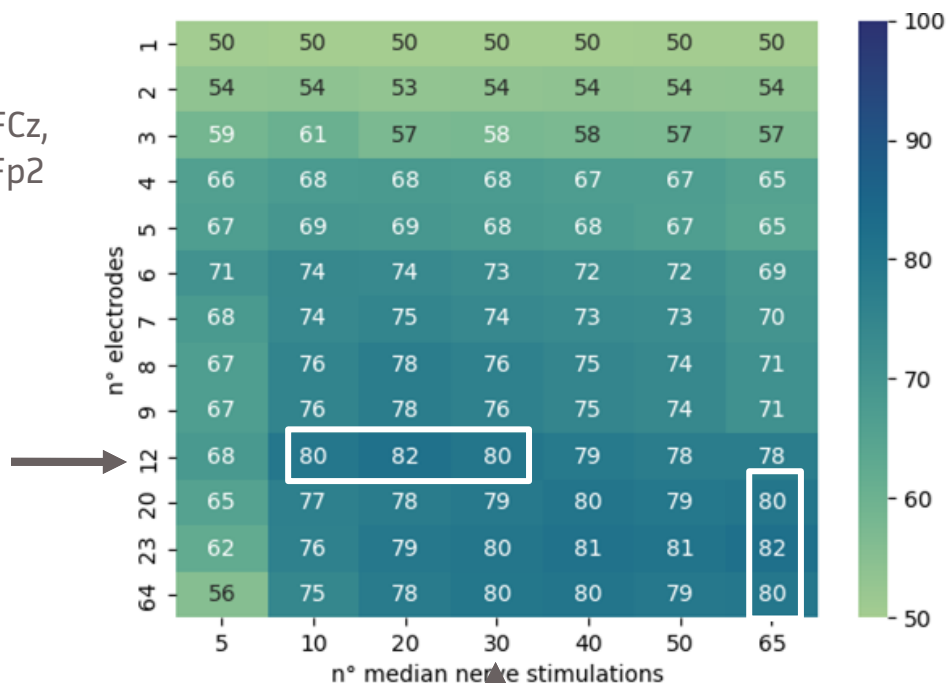
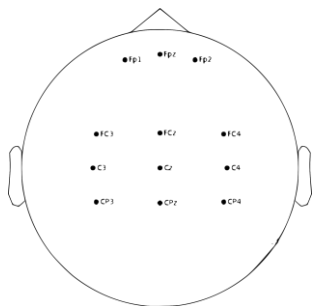
Distance to the **MNS-awake centroid** evolves gradually with the depth of anesthesia



Less electrodes and less stimulations?

Test balanced accuracy in the **MNS-awake** vs **MNS-anesthesia** classification:

C3, C4, Cz, FC4, FC3, FCz,
CP4, CP3, CPz, Fp1, Fpz, Fp2



Future clinical protocol: comparison with other monitor

OBJECTIVE: Compare OCRMDM index with Patient State Index (PSI) by SedLine with 45 subjects
Submitted July 4th



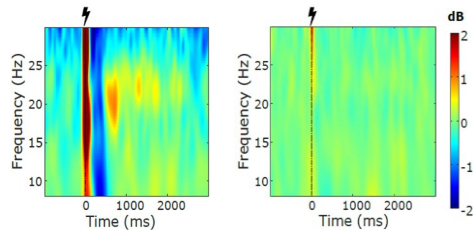
SedLine monitor

Summary: MNS to monitor anesthesia

- ERD/ERS% patterns are no longer visible at deeper levels of anesthesia.
- But they evolve gradually with increasing anesthetic depth.
- A One-Class Riemannian Minimum Distance to the Mean algorithm trained only on awake data correctly differentiates awake vs. anesthetized states (82% when trained with 12 electrodes and 20 stimulations).

Can movement intentions still be detected under anesthesia
using **other MNS-related EEG features**?

Towards **multimodal** movement detection using MNS under general anesthesia



2025 IEEE INTERNATIONAL WORKSHOP ON MACHINE LEARNING FOR SIGNAL PROCESSING, AUG. 31– SEP. 3, 2025, ISTANBUL, TURKEY

RIEMANNIAN FUSIONS OF EEG-BASED FEATURES FOR MOTOR IMAGERY DETECTION UNDER PROPOFOL SEDATION

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Camilla Mannino[‡]
Sébastien Rimbart^{*}

Marie-Constance Corsi[‡]
Laurent Bougrain^{†‡}

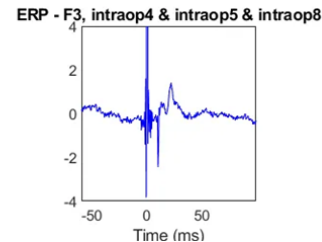
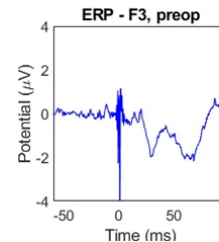
Covariance matrices

Connectivity matrices

NEW EEG FEATURES

Spectral entropy

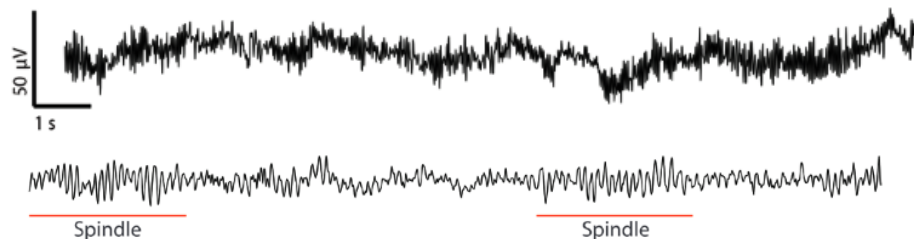
Somatosensory evoked potentials



Discussion

Can we use EEG patterns induced by MNS to detect **other consciousness states**?

- Propofol anesthesia
- Other anesthetics (ketamine, dexmedetomidine...) [17]
- Locked-in state
- Coma
- Hypnosis [20]
- Sleep (dreaming, not dreaming)



Hypnotic State Modulates Sensorimotor Beta Rhythms During Real Movement and Motor Imagery

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¹ Université de Lorraine, CNRS, Inria, LORIA, Nancy, France, ² Independent Researcher, Dambach-la-ville, France,

³ Hemodialysis Department, University Hospital of Strasbourg, Strasbourg, France

[17] Purdon et al, Anesth, 2015.

[20] Rimbart et al, Front Psychology, 2019.

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Associate Professor
Université de Lorraine



Sébastien Rimbart
Inria Starting Faculty Position
Inria Center at the University of Bordeaux

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